WORKSHEET 3.4

Derivatives of Inverse Trigonometric Functions

Unit 3: Composite, Implicit and Inverse Functions

Key ideas

- $\dfrac{d}{dx}[\arcsin x] = \dfrac{1}{\sqrt{1-x^2}}$ and $\dfrac{d}{dx}[\arccos x] = -\dfrac{1}{\sqrt{1-x^2}}$.
- $\dfrac{d}{dx}[\arctan x] = \dfrac{1}{1+x^2}$.
- With an inside function u, replace x by u and multiply by u': $\dfrac{d}{dx}[\arctan u] = \dfrac{u'}{1+u^2}$.
- Square the *whole* inside: for $\arctan(3x)$ the denominator is $1+9x^2$.

Worked examples

Example 1 · With the chain rule

Find $\dfrac{d}{dx}[\arcsin(2x)]$.

Solution. $\dfrac{1}{\sqrt{1-(2x)^2}} \cdot 2$.

$\dfrac{2}{\sqrt{1-4x^2}}$.

Example 2 · With the product rule

Find $\dfrac{d}{dx}[x \arccos x]$.

Solution. $(1)\arccos x + x \cdot \left(-\dfrac{1}{\sqrt{1-x^2}}\right)$.

$\arccos x - \dfrac{x}{\sqrt{1-x^2}}$.

Example 3 · A tangent line

Write an equation of the line tangent to $y = \arccos x$ at $x = \dfrac{1}{2}$.

Solution. The point is $\left(\dfrac{1}{2}, \dfrac{\pi}{3}\right)$. The slope is $-\dfrac{1}{\sqrt{1-\frac{1}{4}}} = -\dfrac{2}{\sqrt{3}}$.

$y - \dfrac{\pi}{3} = -\dfrac{2}{\sqrt{3}}\left(x - \dfrac{1}{2}\right)$.

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Practice

CORE

1. Find $\frac{d}{dx}[4\arctan x]$.

2. Find $\frac{d}{dx}[\arccos(3x)]$.

3. Find $\frac{d}{dx}[\arctan(5x)]$.

4. Find the slope of the graph of $y = \arctan x$ at $x = 2$.

EXAM LEVEL

5. Find $\frac{d}{dx}[\arcsin(e^x)]$.

6. Find $\frac{d}{dx}\left[\arctan\left(\frac{1}{x}\right)\right]$ in simplified form.

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7. Find $\dfrac{d}{dx}\left[(\arcsin x)^2\right]$.

8. Let $f(x) = \arctan\left(\dfrac{x}{3}\right)$. Find $f'(3)$.

Question 9 · Free-response practice

Let $f(x) = \arctan(2x)$.

(a) Find $f'(x)$.

(b) Write an equation of the line tangent to the graph of f at $x = \dfrac{1}{2}$.

(c) Find all values of x at which the slope of the graph of f is $\dfrac{2}{5}$.

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Worked solutions

1. $\dfrac{4}{1+x^2}$.

2. $-\dfrac{3}{\sqrt{1-9x^2}}$.

3. $\dfrac{5}{1+25x^2}$.

4. $\dfrac{1}{1+4} = \dfrac{1}{5}$.

5. $\dfrac{e^x}{\sqrt{1-e^{2x}}}$.

6. $\dfrac{1}{1+\frac{1}{x^2}} \cdot \left(-\dfrac{1}{x^2}\right) = -\dfrac{1}{x^2+1}$.

7. $\dfrac{2\arcsin x}{\sqrt{1-x^2}}$.

8. $f'(x) = \dfrac{1/3}{1+x^2/9}$, so $f'(3) = \dfrac{1/3}{2} = \dfrac{1}{6}$.

9. (a) $f'(x) = \dfrac{2}{1+4x^2}$.

   (b) $f\left(\dfrac{1}{2}\right) = \arctan 1 = \dfrac{\pi}{4}$ and $f'\left(\dfrac{1}{2}\right) = 1$: $y - \dfrac{\pi}{4} = x - \dfrac{1}{2}$.

   (c) $\dfrac{2}{1+4x^2} = \dfrac{2}{5}$ gives $1+4x^2 = 5$, so $x = \pm 1$.

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